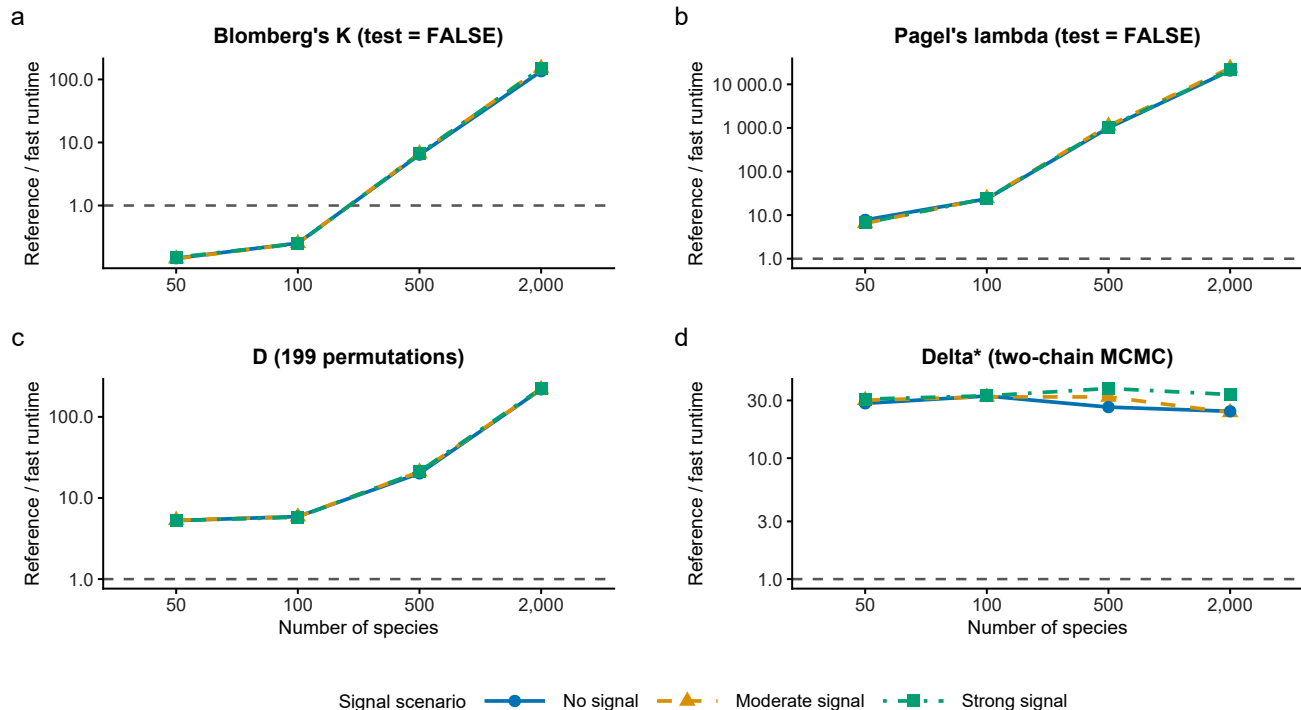


Matched single-call runtime: fastphylosig 0.3.0 vs reference implementations

Reference / fastphylosig ratio of median wall-clock time; horizontal rule marks parity.



Speedup = median(reference) / median(fastphylosig); >1 favours fastphylosig, <1 favours reference.

Points summarize 10 paired one-call timings per cell; tree, trait, and analysis-context preparation was excluded.

K/lambda use test=FALSE. D compares fast_d(test=TRUE, nsim=199) with caper::phylo.d(permut=199).

Delta uses two MCMC chains (10,000 iterations/chain, thin=10, burn=100) and is compared with the pinned delta() reference.

Delta caveat: reference warnings in 112/120 calls (NaNs, Inf replacement, complex coercion, non-finite gradients); fast Delta had one optimizer-limit warning. The reference returns no convergence object.

Runtime only; no equal-accuracy or convergence claim. X positions are categorical/equally spaced; each panel has its own logarithmic y-axis.